

ECE 461 - HW01

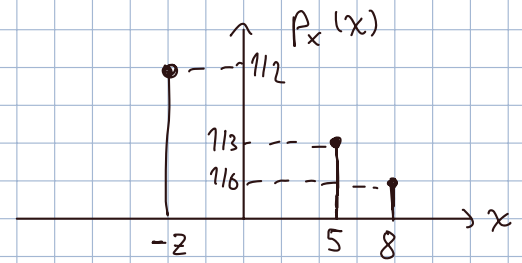
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1.a) Roll ① with probability $\frac{1}{6} \rightarrow \text{Win } 8\$ \Leftrightarrow X=8$

Roll ②, ③ with probability $\frac{2}{6} \rightarrow \text{Win } 5\$ \Leftrightarrow X=5$

Roll ④, ⑤, ⑥ with probability $\frac{3}{6} \rightarrow \text{Lose } z\$ \Leftrightarrow X=-z$

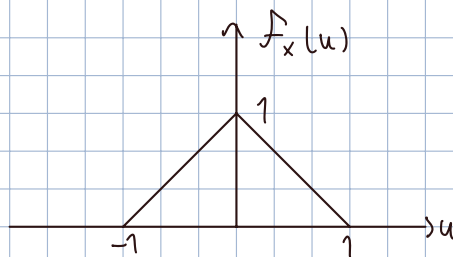
$$P_X(x) = P(X=x) = \begin{cases} 1/6, & x=8 \\ 1/3, & x=5 \\ 1/2, & x=-z \\ 0, & \text{else} \end{cases}$$



1.b) $\mu_x = \sum_{x \in S_x} x P_X(x) = \frac{1}{6} \cdot 8 + \frac{1}{3} \cdot 5 - \frac{1}{2} \cdot z = -1 \Rightarrow \boxed{z = 8}$

1.c) $E[\ln |X|] = \sum_{x \in S_x} \ln |x| P_X(x) = \frac{1}{6} \ln 8 + \frac{1}{3} \ln 5 + \frac{1}{2} \ln 8$
 $\approx \boxed{1.9228}$

$$2.) \quad f_x(u) = \begin{cases} 1-|u| & , \quad 0 \leq |u| \leq 1 \\ 0 & , \quad \text{else} \end{cases}$$



$$2.a) \quad P\{X > 1/2\} = \int_{1/2}^{\infty} f_x(u) du = \int_{1/2}^1 (1-u) du = u - \frac{u^2}{2} \Big|_{1/2}^1 \\ = 1 - \frac{1}{2} - \left(\frac{1}{2} - \frac{1}{8}\right) = \boxed{\frac{1}{8}}$$

$$2.b) \quad P\{X > 0 \mid X < 1/2\} = \frac{P\{0 < X < 1/2\}}{P\{X < 1/2\}} \rightarrow P\{X > 0\} = \frac{1}{2} \quad \text{since pdf is symmetric around 0.}$$

Then $P\{0 < X < \frac{1}{2}\} = P\{X > 0\} - P\{X > \frac{1}{2}\}$

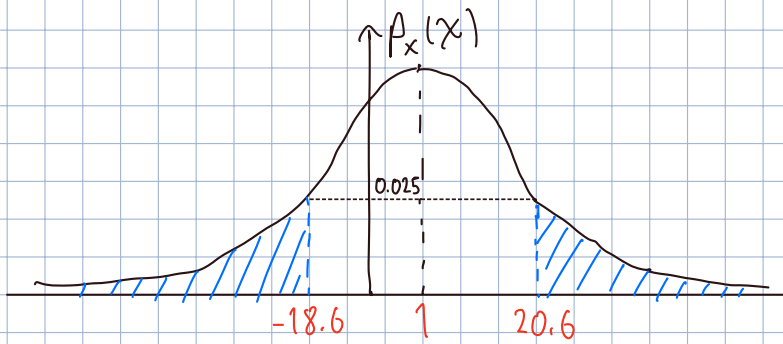
$$= \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$$

↓

$$1 - P\{X > \frac{1}{2}\} = 1 - \frac{1}{8} = \frac{7}{8}$$

$$P\{X > 0 \mid X < \frac{1}{2}\} = \frac{3/8}{7/8} = \boxed{\frac{3}{7}}$$

3) $X \sim \mathcal{N}(\mu, \sigma^2)$ with $P\{X > 20.6\} = P\{X \leq -18.6\} = 0.025$



→ 2 points with same probability value on Gaussian pdf must be symmetric around the mean (equally spaced from the mean)

↳ Then mean is located in the middle of those points } $\mu = \frac{20.6 - 18.6}{2} = \boxed{1}$

→ Standardizing X and checking the cumulative distribution function (CDF) at -18.6

$$P\{X < -18.6\} = \Phi\left(\frac{-18.6 - \mu}{\sigma}\right) = \Phi\left(\frac{-19.6}{\sigma}\right) = 0.025$$

Using "norminv" function of the MATLAB, it's found that standard normal random variable has CDF 0.025 at -1.96

↳ $\text{norminv}(0.025) = -1.96 = \frac{-19.6}{\sigma} \Rightarrow \boxed{\sigma = 10}$

$$4.a) \quad Q(a) = \int_a^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = \frac{1}{\sqrt{2\pi}} \int_a^{\infty} \underbrace{\frac{1}{t}}_u \underbrace{t e^{-\frac{t^2}{2}}}_{dV} dt$$

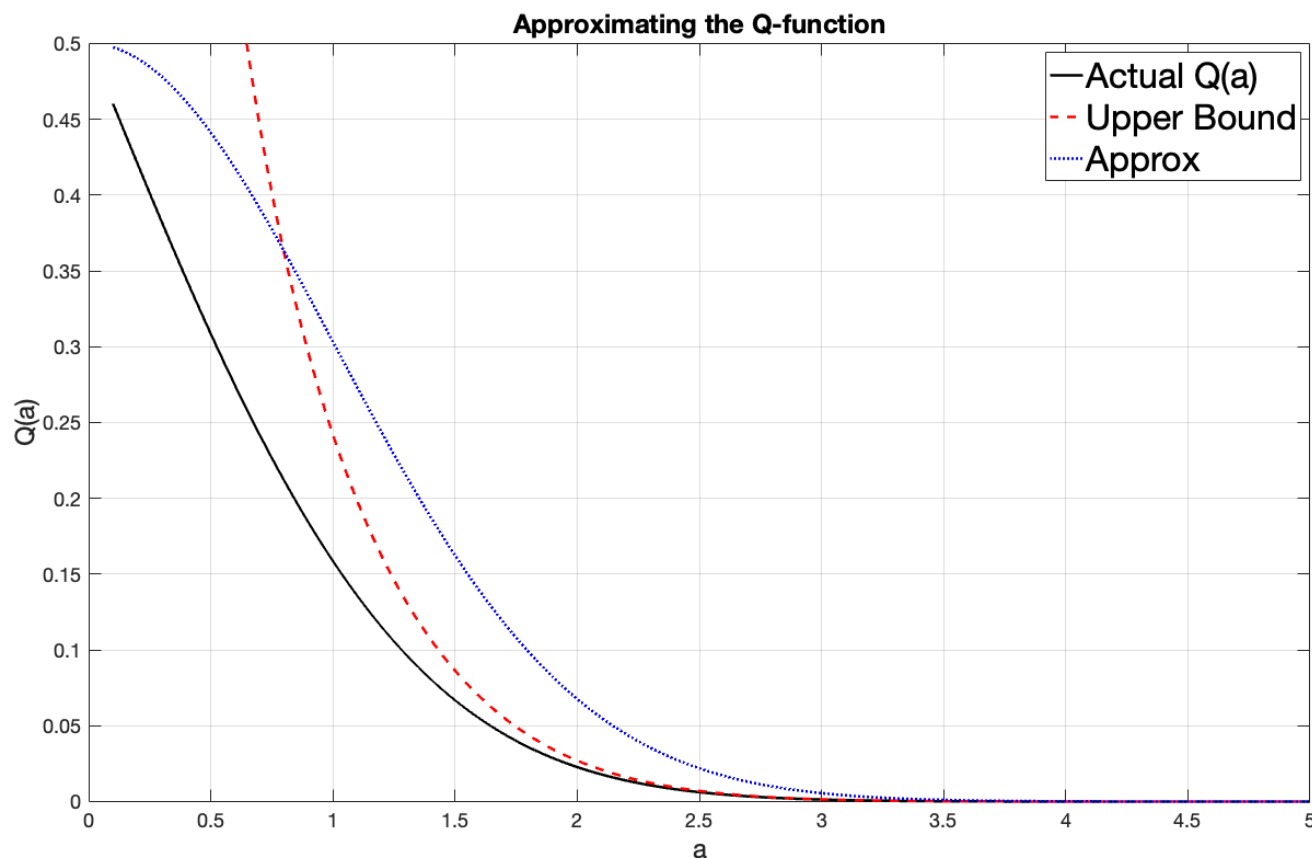
→ Notice that $\frac{d}{dt} [e^{-\frac{t^2}{2}}] = -t e^{-\frac{t^2}{2}}$. Then $V = -e^{-\frac{t^2}{2}}$ and $du = -\frac{1}{t^2}$

$$Q(a) = -\frac{1}{\sqrt{2\pi}} \left[\frac{1}{t} e^{-\frac{t^2}{2}} \right]_a^{\infty} - \frac{1}{\sqrt{2\pi}} \int_a^{\infty} \frac{1}{t^2} e^{-\frac{t^2}{2}} dt$$

Integrand is always positive (or zero)
and thus integral is always non-negative
 ≥ 0

$$Q(a) \leq \frac{1}{\sqrt{2\pi}} \left[\frac{1}{a} e^{-\frac{a^2}{2}} - \underbrace{\lim_{t \rightarrow \infty} \frac{1}{t} e^{-\frac{t^2}{2}}}_0 \right] \Rightarrow Q(a) \leq \frac{1}{a\sqrt{2\pi}} e^{-\frac{a^2}{2}}$$

4.b



→ As can be seen from the plot, upper bound is always larger than $Q(a)$, even more so for larger a . Also exponential approximation is decately close to $Q(a)$ especially (again) for large a . But there is a significant difference for $a < 2$

$$5.a) \quad F_X(x) = [1 - c e^{-x/d}] u(x)$$

For a valid CDF, one must have:

$$1) \lim_{x \rightarrow -\infty} F_X(x) = 0 \Rightarrow \text{In our case, since } \lim_{x \rightarrow -\infty} u(x) = 0, \text{ this is always satisfied}$$

$$2) \lim_{x \rightarrow \infty} F_X(x) = 1 \Rightarrow \lim_{x \rightarrow \infty} F_X(x) = 1 - c \lim_{x \rightarrow \infty} e^{-x/d} \Rightarrow \text{we require } \underline{d > 0}$$

0 if $d > 0$, diverges otherwise

$$3) 0 \leq F_X(x) \leq 1 \Rightarrow \text{For } d > 0, 0 \leq e^{-x/d} \leq 1, \text{ Then we need } \underline{0 < c \leq 1}$$

We require $d > 0$ and $0 < c \leq 1$ for $F_X(x)$ to be valid CDF.

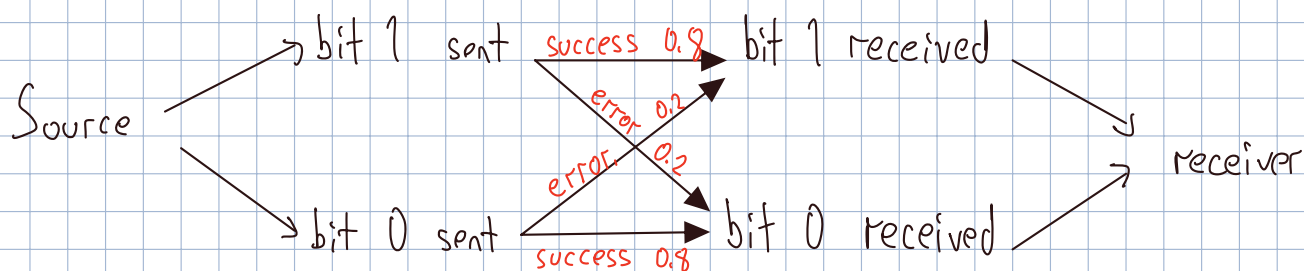
* Note that $F_X(x)$ is continuous and monotonically increasing if these conditions are satisfied.

$$5.b) \quad f_X(x) = k e^{-ax} u(x)$$

A valid PDF must be normalized:

$$1 = \int_{-\infty}^{\infty} f_X(x) dx = \int_0^{\infty} k e^{-ax} dx = \frac{k}{a} \left[e^{-ax} \right]_0^{\infty} = \frac{k}{a} = 1 \Rightarrow \boxed{k = a}$$

6.a)



Let S_i / R_i be the events of sending/receiving bit $i \in \{0, 1\}$

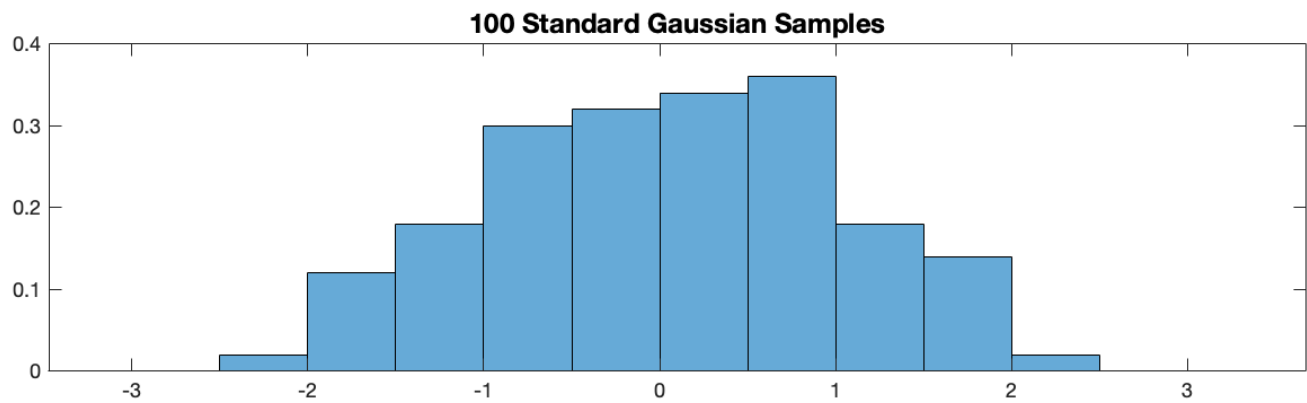
$$P(R_1) = \underbrace{P(R_1|S_1)}_{\text{Success } 0.8} \underbrace{P(S_1)}_{0.7} + \underbrace{P(R_1|S_0)}_{\text{(error) } 0.2} \underbrace{P(S_0)}_{0.3} = 0.56 + 0.06 = \boxed{0.62}$$

6.b) $P(S_1|R_1) = ?$

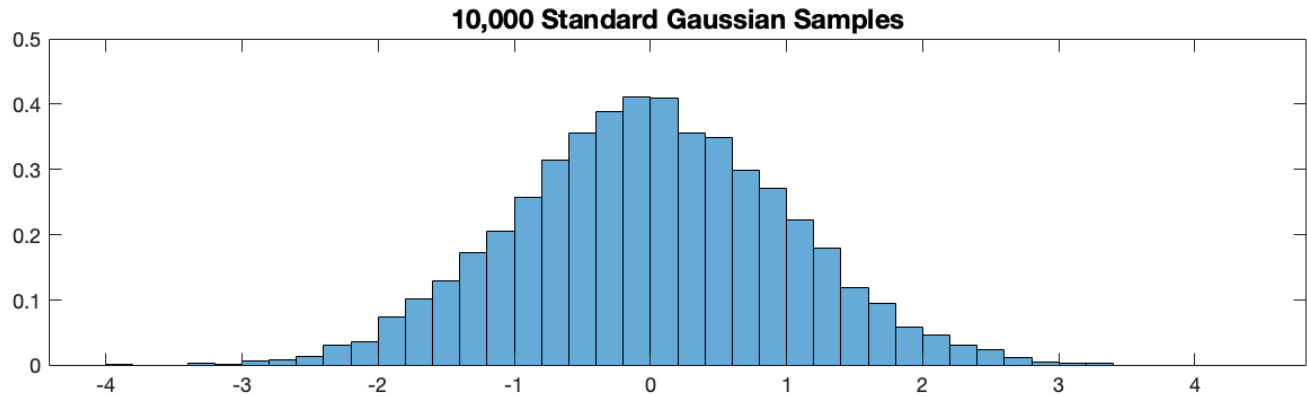
Using Bayes Rule

$$P(S_1|R_1) = \frac{\overbrace{P(R_1|S_1)}^{0.8} \overbrace{P(S_1)}^{0.7}}{\underbrace{P(R_1)}_{\text{found as } 0.62 \text{ from previous options}}} = \frac{0.8 \times 0.7}{0.62} = 0.903$$

7.a)

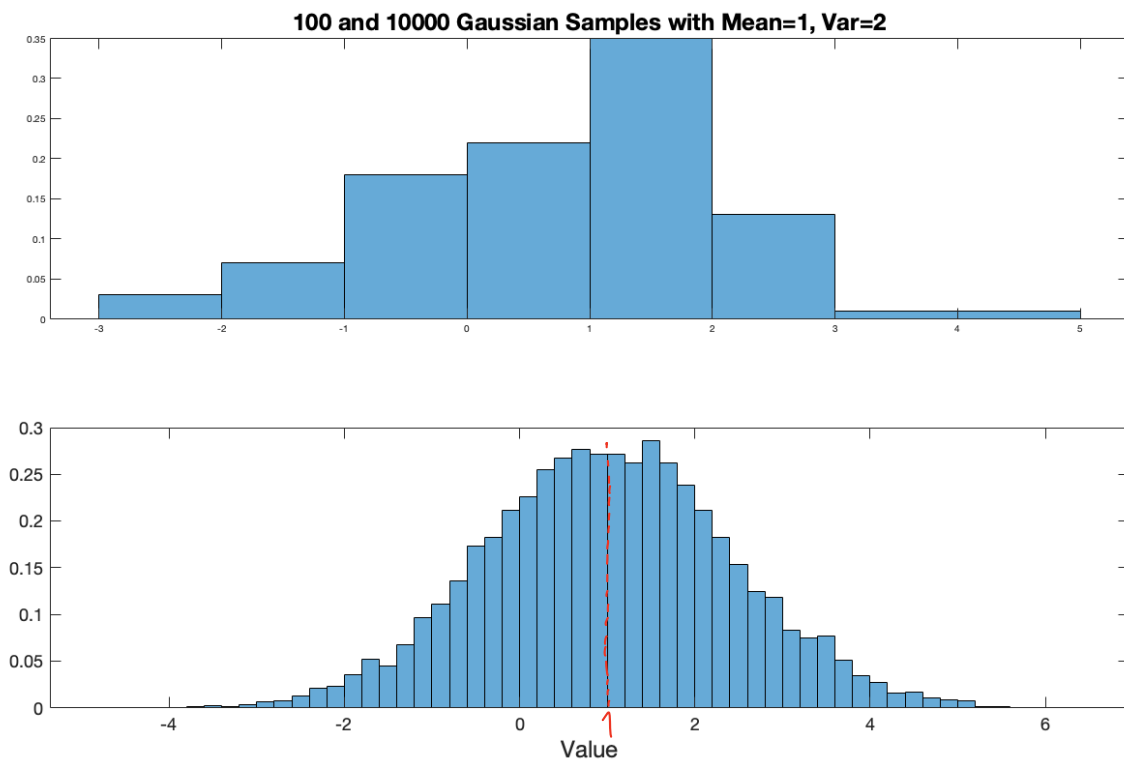


7.b)



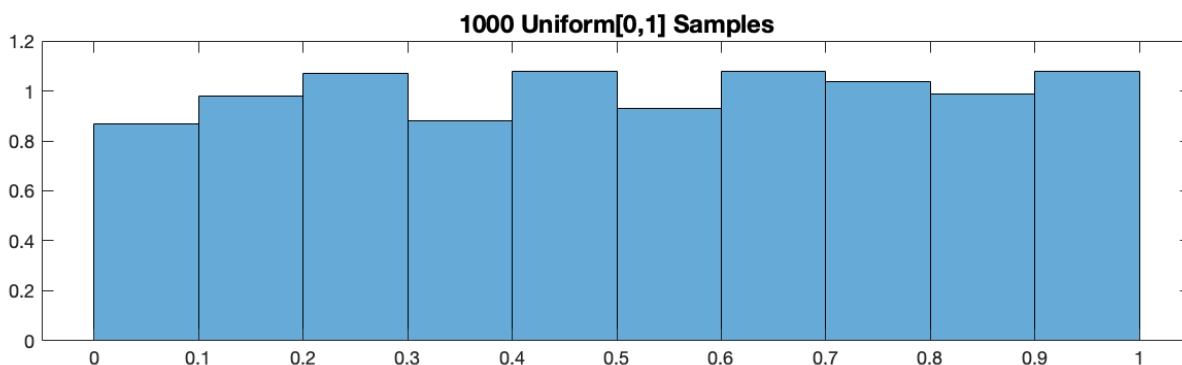
* With only 100 samples, histogram hardly resembles to a Gaussian. Increasing sample size to 10 000 makes the underlying normal distribution. We observe that mean is indeed around 0 and standard deviation is around 1. Spread of PDF (largest nonzero value) is around 3 for both PDFs. Clearly, 100 samples are not enough for approximating the Gaussian while 10 000 samples performs quite well.

7.c)



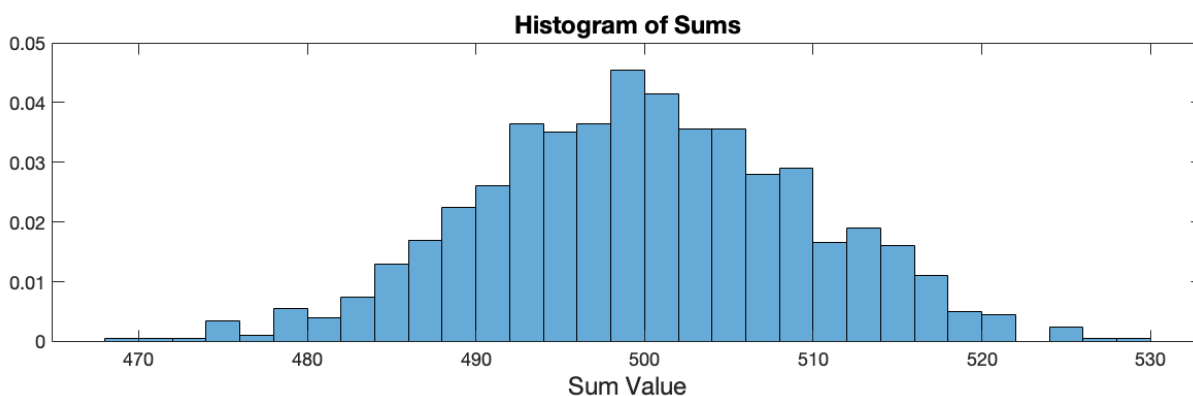
As in a, b; 10 000 samples approximate the Gaussian much better compared to 100. In this case, it's apparent that mean has shifted to 1 and standard deviation is higher since the spread of the PDF is larger now.

7.d)



→ Histogram is close to uniform. It seems like 1000 samples are not still quite enough

7.e)



Central limit theorem claims that sum of large number of random variables (regardless of their individual distribution) will converge to a Gaussian as # of summed R.V.s approach to infinity

As expected, histogram resembles to a Gaussian random variable. Also CLT claims

$$\mu_{\text{sum}} = \sum_{i=1}^N \mu_i$$

With each uniform R.V. having a mean of $\frac{1}{2}$, we observe that mean of the sum is $1000 \times \frac{1}{2} = 500$ as expected

7.e)

MATLAB CODE

```
1 % Parts a,b
2 figure;
3 subplot(2,1,1);
4 data100 = randn(1, 100);
5 histogram(data100, 'Normalization', 'pdf', 'BinWidth', 0.5);
6 set(gca, 'FontSize', 18);
7 title('100 Standard Gaussian Samples', 'FontSize', 24);
8
9 subplot(2,1,2);
10 data10000 = randn(1, 10000);
11 histogram(data10000, 'Normalization', 'pdf');
12 set(gca, 'FontSize', 18);
13 title('10,000 Standard Gaussian Samples', 'FontSize', 24);
14
15 % Part c
16 figure;
17 data_custom_100 = 1 + sqrt(2) * randn(1, 100);
18 data_custom_10000 = 1 + sqrt(2) * randn(1, 10000);
19 subplot(2,1,1);
20 histogram(data_custom_100, 'Normalization', 'pdf');
21 title('100 and 10000 Gaussian Samples with Mean=1, Var=2', 'FontSize', 24);
22 subplot(2,1,2);
23 histogram(data_custom_10000, 'Normalization', 'pdf');
24 set(gca, 'FontSize', 18);
25 xlabel('Value', 'FontSize', 24);
26
27 % Parts d, e
28 figure;
29 subplot(2,1,1);
30 uni_data = rand(1, 1000);
31 histogram(uni_data, 'Normalization', 'pdf', 'BinWidth', 0.1);
32 set(gca, 'FontSize', 18);
33 title('1000 Uniform[0,1] Samples', 'FontSize', 24);
34
35
36 subplot(2,1,2);
37 % Generate a 1000x1000 matrix and sum across rows
38 sum_of_uniform_data = sum(rand(1000, 1000), 2);
39 histogram(sum_of_uniform_data, 'Normalization', 'pdf', 'BinWidth', 2);
40 set(gca, 'FontSize', 18);
41 title('Histogram of Sums', 'FontSize', 24);
42 xlabel('Sum Value', 'FontSize', 24);
```

*Note that "Normalization" and "pdf" options were used plotting the histogram.

